PHYC10009 Foundations of Physics – Lecture 8

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Physics, Room 501

(Please switch your mobile phone to silent)

Acceleration - G5.1

Speed is constant

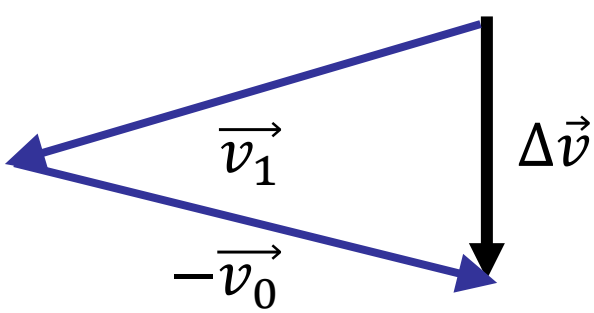
Passenger is accelerating

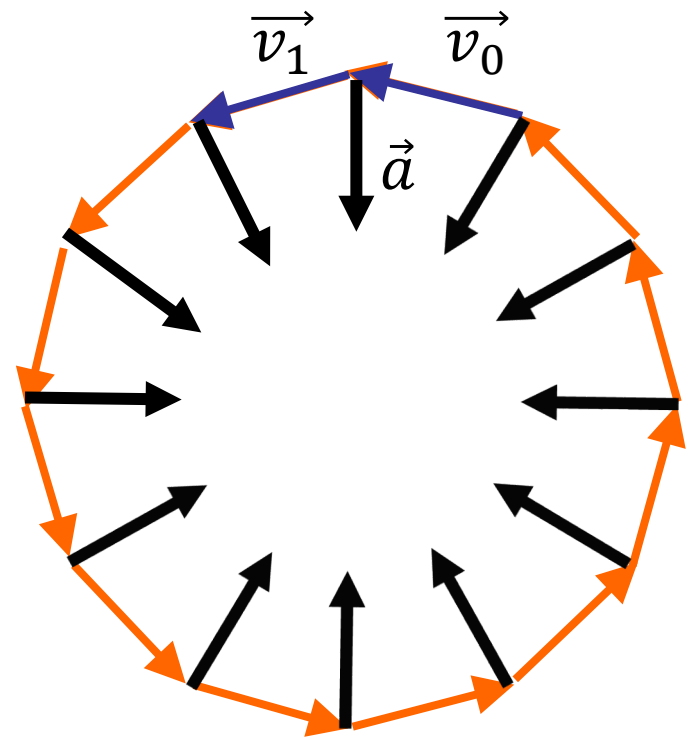
Acceleration directed toward *centre* of motion

Centripetal acceleration

More on this later

$$\Delta\vec{v} = \overrightarrow{v_1} - \overrightarrow{v_0}$$

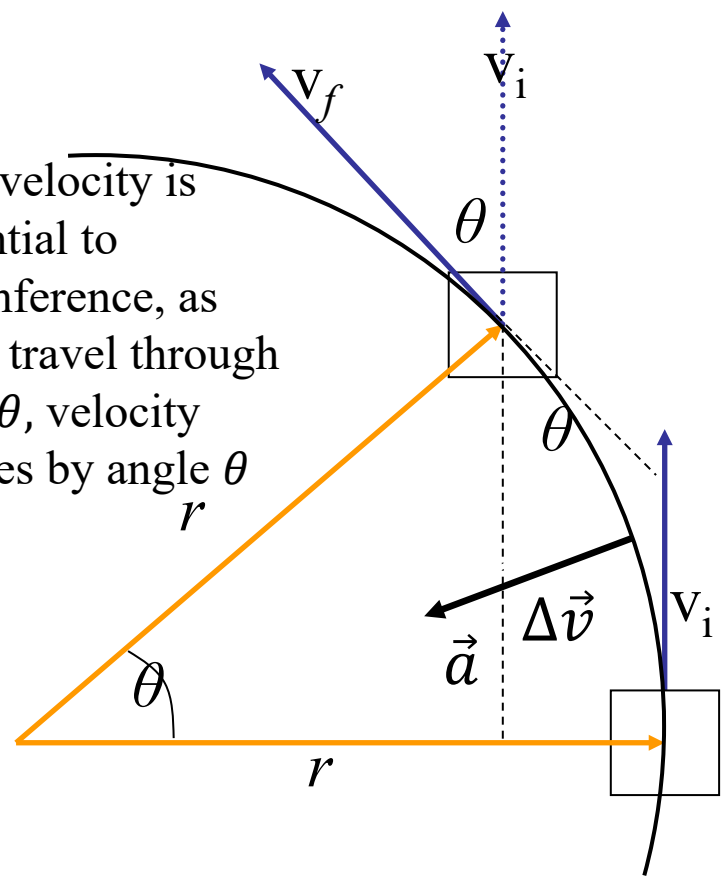




$$\vec{a}_{avg} = \frac{\Delta\vec{v}}{\Delta t} = \frac{\overrightarrow{v_1} - \overrightarrow{v_0}}{\Delta t}$$

Acceleration - G5.1

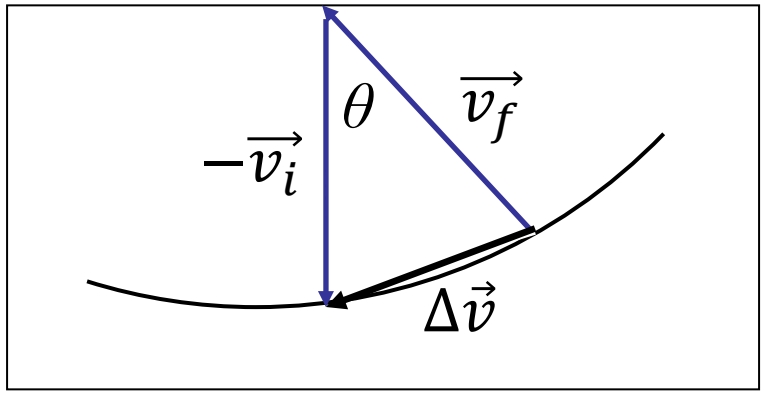
Since velocity is tangential to circumference, as object travel through angle θ, velocity changes by angle θ



Average acceleration over time interval Δt:

$$\vec{a} = \frac{\Delta\vec{v}}{\Delta t}$$

$$\Delta\vec{v} = \vec{v_f} - \vec{v_i}$$

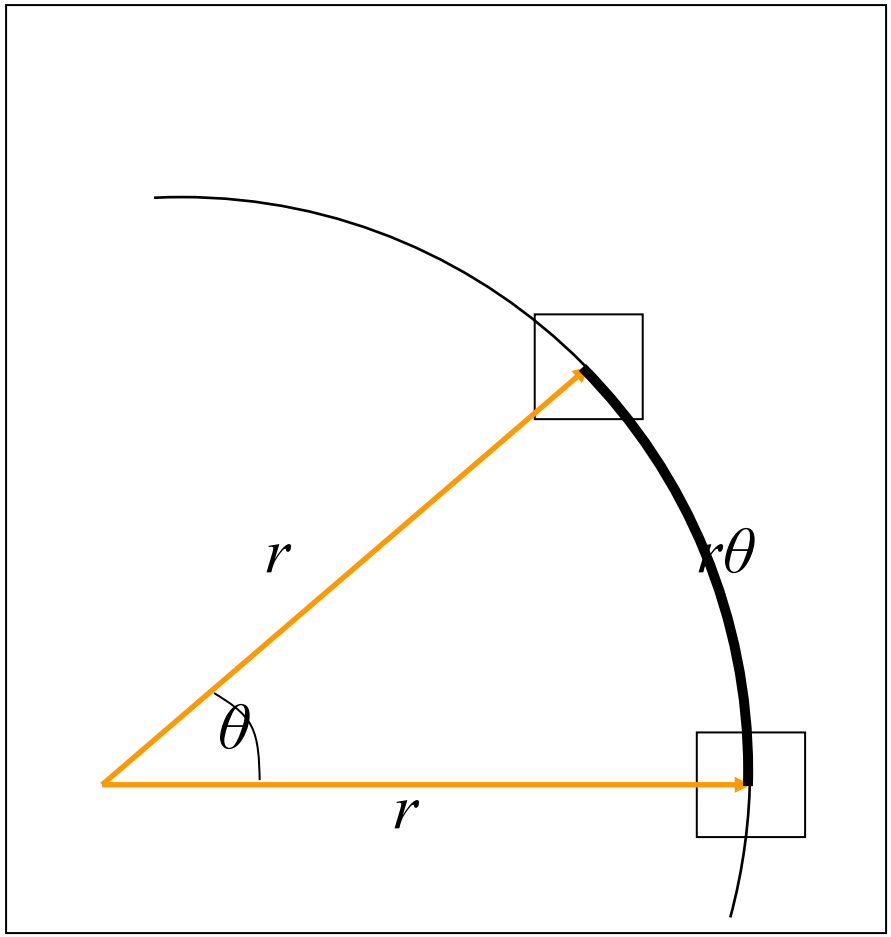


$$|\vec{v_i}| = |\vec{v_f}| = v$$

$$|\Delta\vec{v}| = v\theta$$

Acceleration is directed towards the centre of the circle.

Acceleration – G 5.1



Recall $\vec{a} = \dfrac{\Delta\vec{v}}{\Delta t}$

From previous slide: $|\Delta\vec{v}| = v\theta$

From diagram at left: $v = |\vec{v}| = \dfrac{r\theta}{\Delta t}$

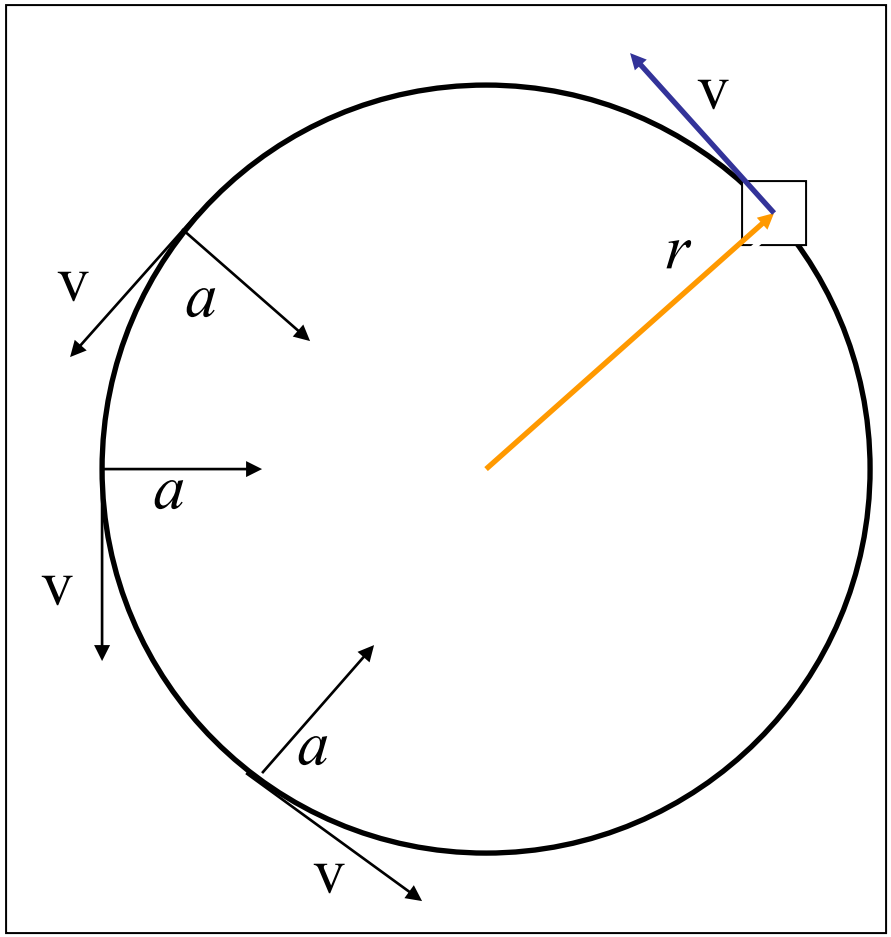
Re-arrange: $\Delta t = \dfrac{r\theta}{\mathrm{v}}$

Use definition: $\overrightarrow{|a|} = \left|\dfrac{\Delta\vec{v}}{\Delta t}\right| = \dfrac{v\theta}{\dfrac{r\theta}{v}}$

Can let Δt go to zero. The magnitude of the *instantaneous* acceleration is given by the equation.

$$\boxed{|\vec{a}| = \frac{v^2}{r}}$$

Summary: uniform circular motion - G 5.1



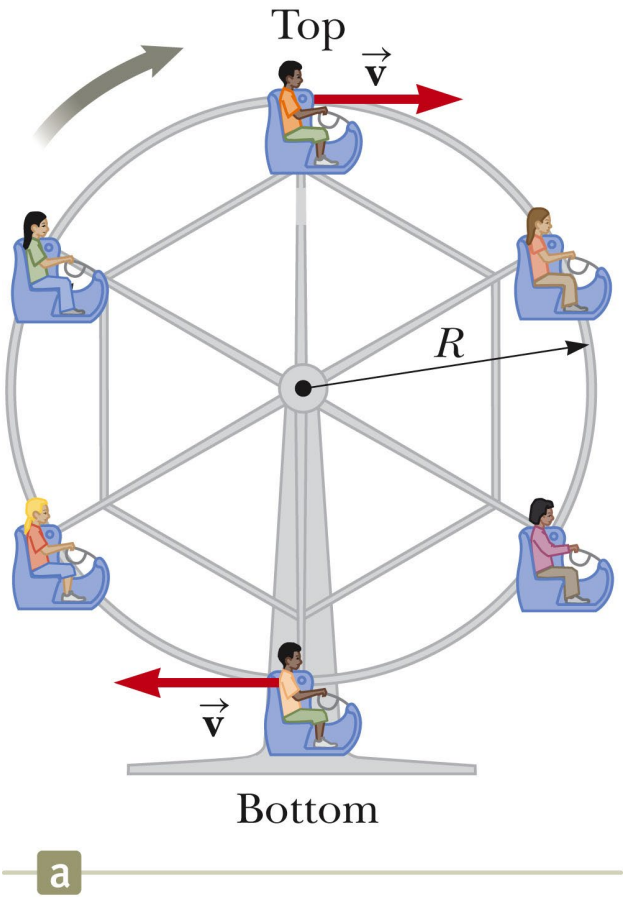
Object in uniform circular motion at constant speed v:

Acceleration is always directed towards the centre of the circle.

Acceleration has constant magnitude:

$$|\vec{a}| = \frac{v^2}{r}$$

Uniform circular motion - G 5.2



For any object in uniform circular motion at constant speed v:

- Acceleration is always directed towards the centre of the circle.
- Acceleration has constant magnitude.

$$\vec{a} = \left(\frac{v^2}{r}, towards\ centre\ of\ circle\right)$$

centripetal acceleration

Newton II: $\vec{F}_{net} = m\vec{a}$

$$\vec{F}_{net} = \left(\frac{mv^2}{r}, towards\ centre\ of\ circle\right)$$

centripetal (net) Force

Centripetal force – G 5.2

$$\vec{F}_{net} = \left(\frac{mv^2}{r}, towards\ centre\ of\ circle\right)$$

centripetal (net) Force

- ‘centripetal’ means ‘centre-seeking’.
- Centripetal force is *not* a new kind of force. It is simply the *net force* that is required for an object to travel in a circle.
- This net force can be due to a single force acting on an object, or a combination of forces acting simultaneously.

Hammer Throw - G5.2

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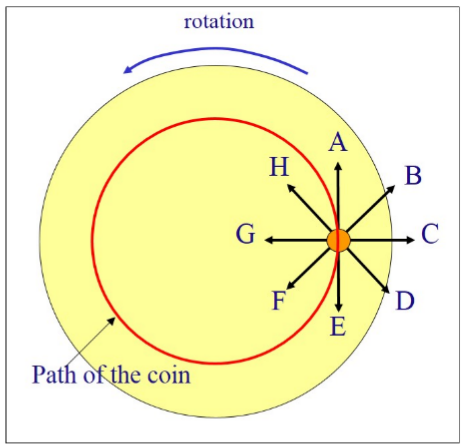
Turntable Friction direction

A coin sits on a rotating turntable.

- What is the direction of the frictional force on the coin?



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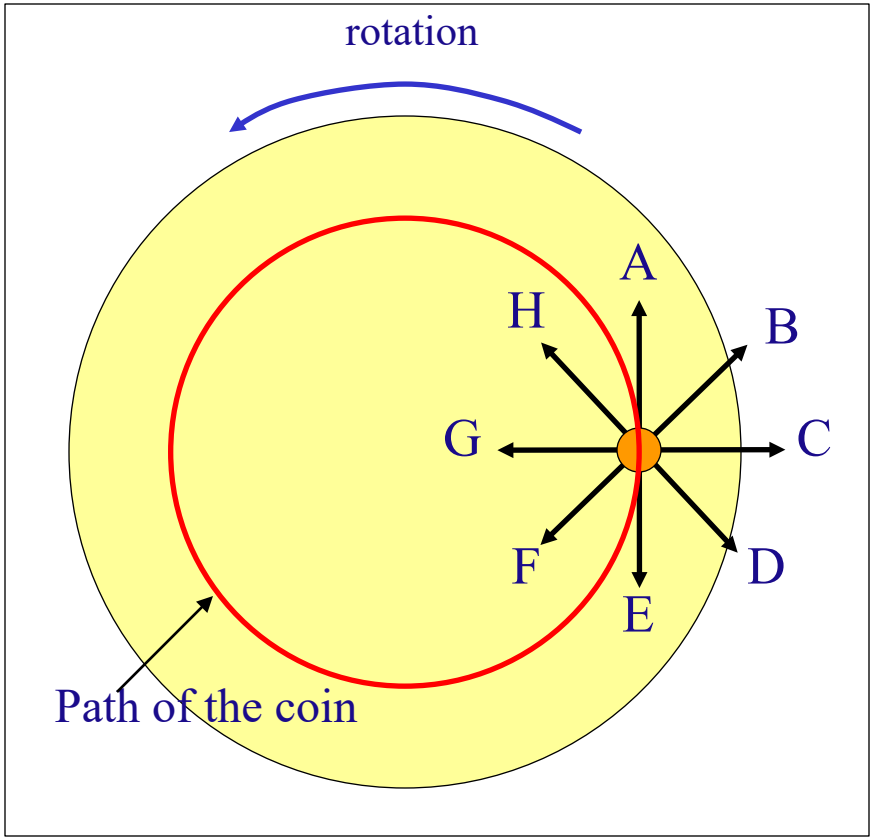
Question - SJWW 5.3

A coin sits on a rotating turntable.

- What is the direction of the frictional force on the coin?

G



D Car turntable

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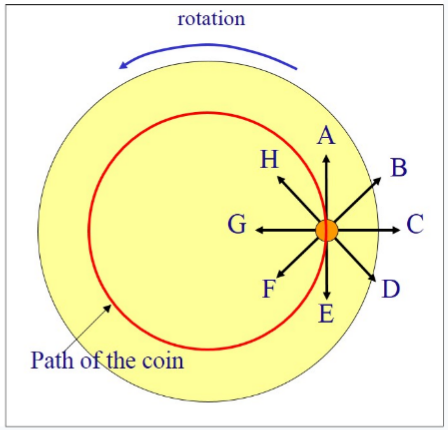
Turntable Friction coin direction

A coin sits on a rotating turntable.

- If friction was suddenly 'switched off', in which direction would the coin move?



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Question - SJWW 5.3

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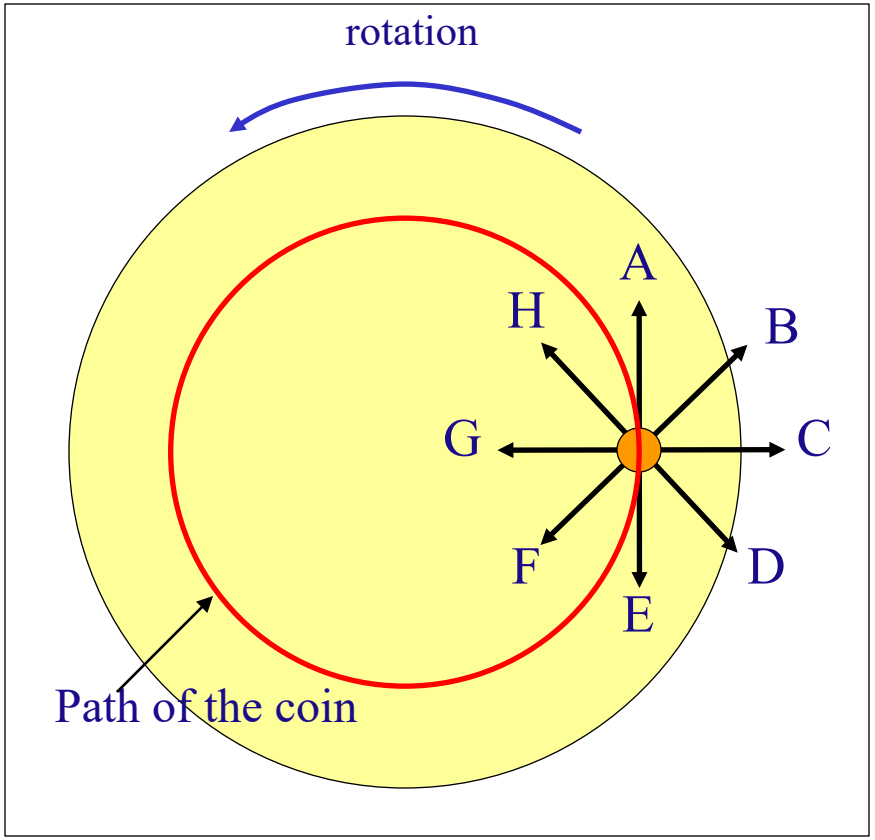
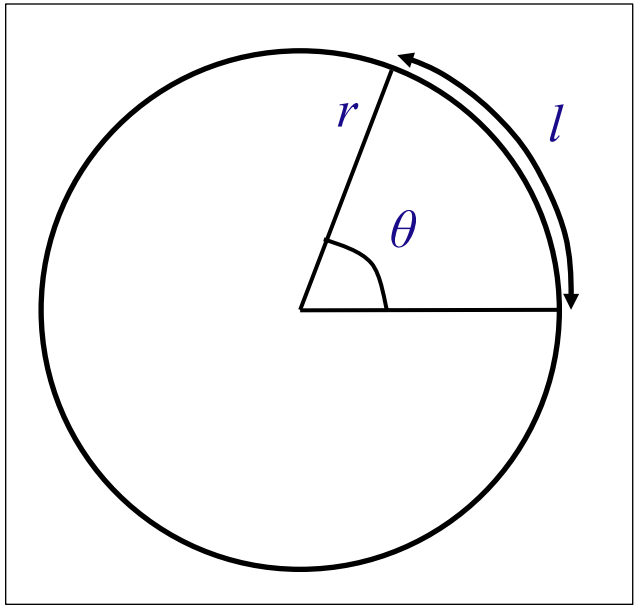
- If friction was suddenly ‘switched off’, in which direction would the coin move?

A



Measuring angles – radians - G 8.1



Circumference of circle = $2\pi r$

$$\frac{\theta}{\text{total angle in a circle}} = \frac{l}{2\pi r}$$

- Definition of radian: Total angle in a circle is 2π radians.

$$\frac{\theta \text{ radian}}{2\pi \text{ radian}} = \frac{l}{2\pi r}$$

- Definition of degree: Total angle in a circle is 360 degrees.

$$\frac{\theta^\circ}{360^\circ} = \frac{l}{2\pi r}$$

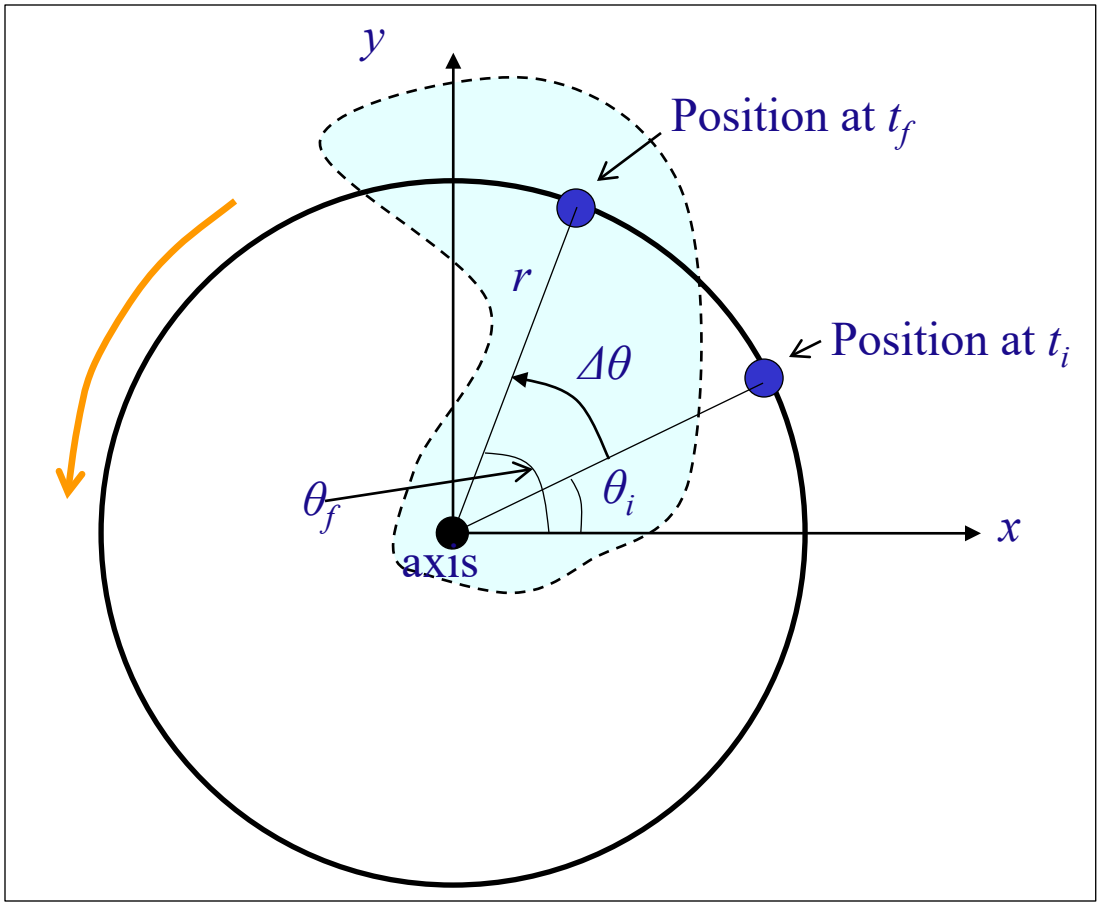
- If θ is expressed in radians, arc length formula is:

$$l = r\theta$$

- Conversion: 2π rad = 360°
  - (1 rad = 57.3°)

Rotational variables - SJWW 9.1

Motion of a point on a rotating object



Angular Position:

θ

Angular displacement:

$\Delta\theta$

Angular speed:

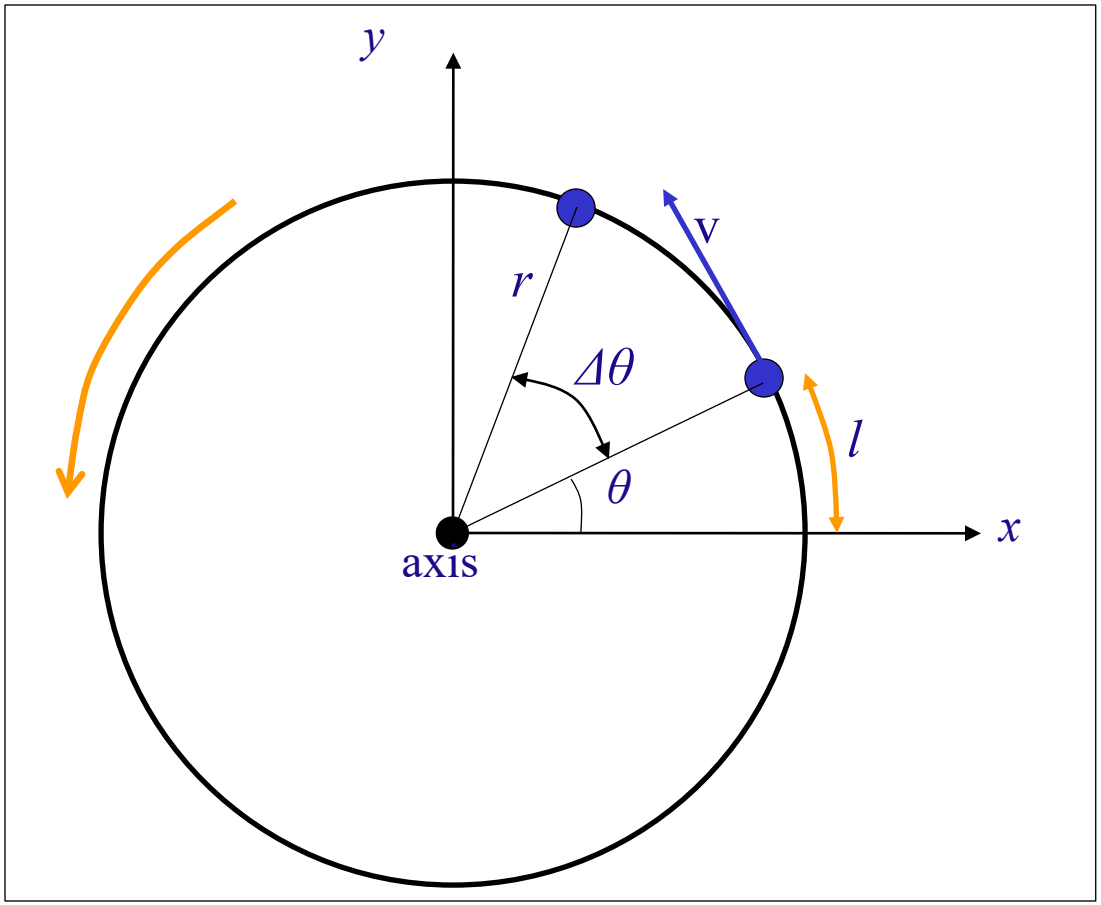
$$\omega_{av} \equiv \frac{\Delta\theta}{\Delta t} \qquad \omega \equiv \frac{d\theta}{dt}$$

Angular acceleration:

$$\alpha_{av} \equiv \frac{\Delta\omega}{\Delta t} \qquad \alpha \equiv \frac{d\omega}{dt}$$

Angular speed - SJWW 9.1

Motion of a point on a rotating object



Angular speed:

$$\omega_{av} \equiv \frac{\Delta\theta}{\Delta t} \qquad \omega \equiv \frac{d\theta}{dt}$$

Units of ω are rad/s.

If the point rotates at constant speed:

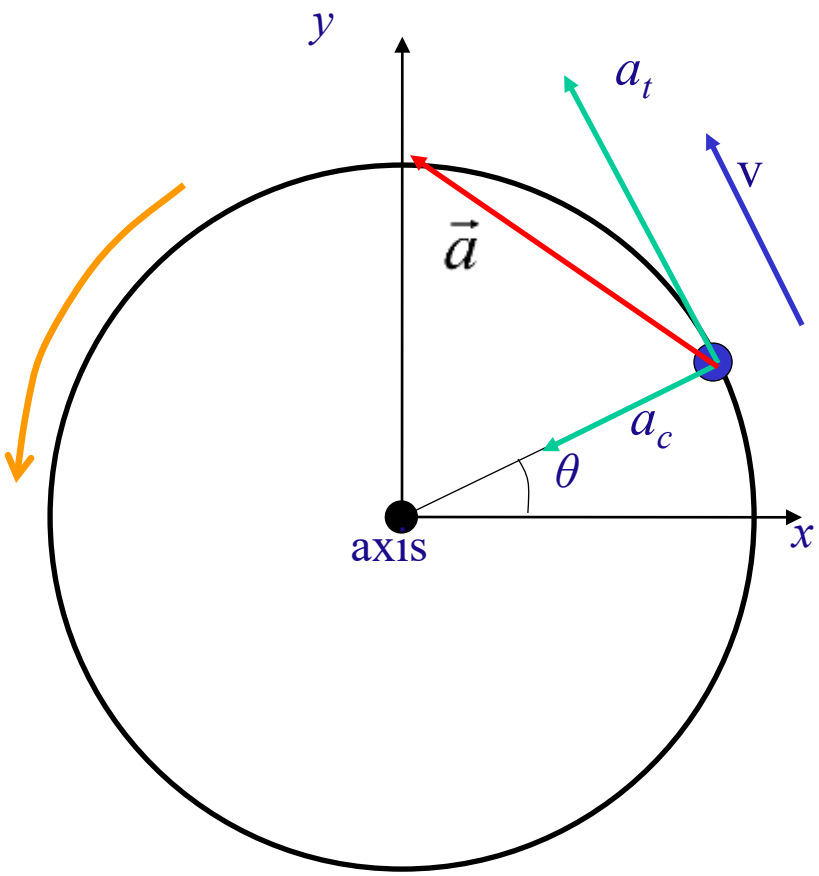
$$v = \frac{2\pi r}{T} \qquad \omega \equiv \frac{\Delta\theta}{\Delta t} = \frac{2\pi}{T}$$

More generally:

$$l = r\theta \qquad \frac{dl}{dt} = r\frac{d\theta}{dt}$$

$$\boxed{v = \omega r}$$

Tangential acceleration - SJWW 3.6



Point speeding up as it moves in a circle

- In *non-uniform circular motion,* the point still moves in a circle, but *also* speeds up or slows down in the direction of motion.

$$v = \omega r$$

$$\frac{dv}{dt} = \left(\frac{d\omega}{dt}\right) r$$

$$a \equiv \frac{dv}{dt} \qquad \alpha \equiv \frac{d\omega}{dt}$$

$$a_t = \alpha r$$

Tangential acceleration

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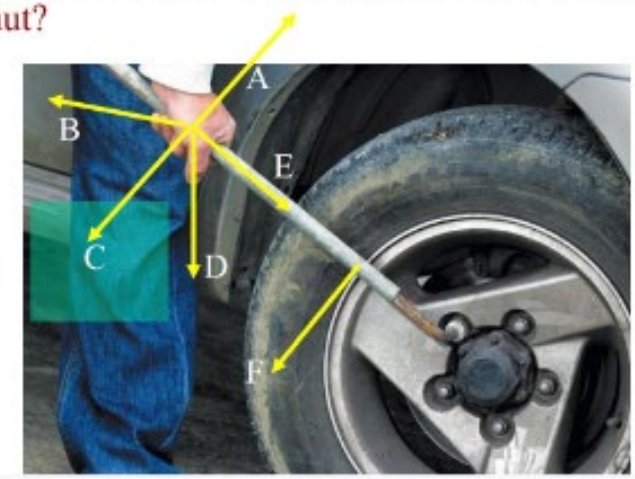
Best way to undo nut?

Which force would be most effective in loosening the wheel nut?



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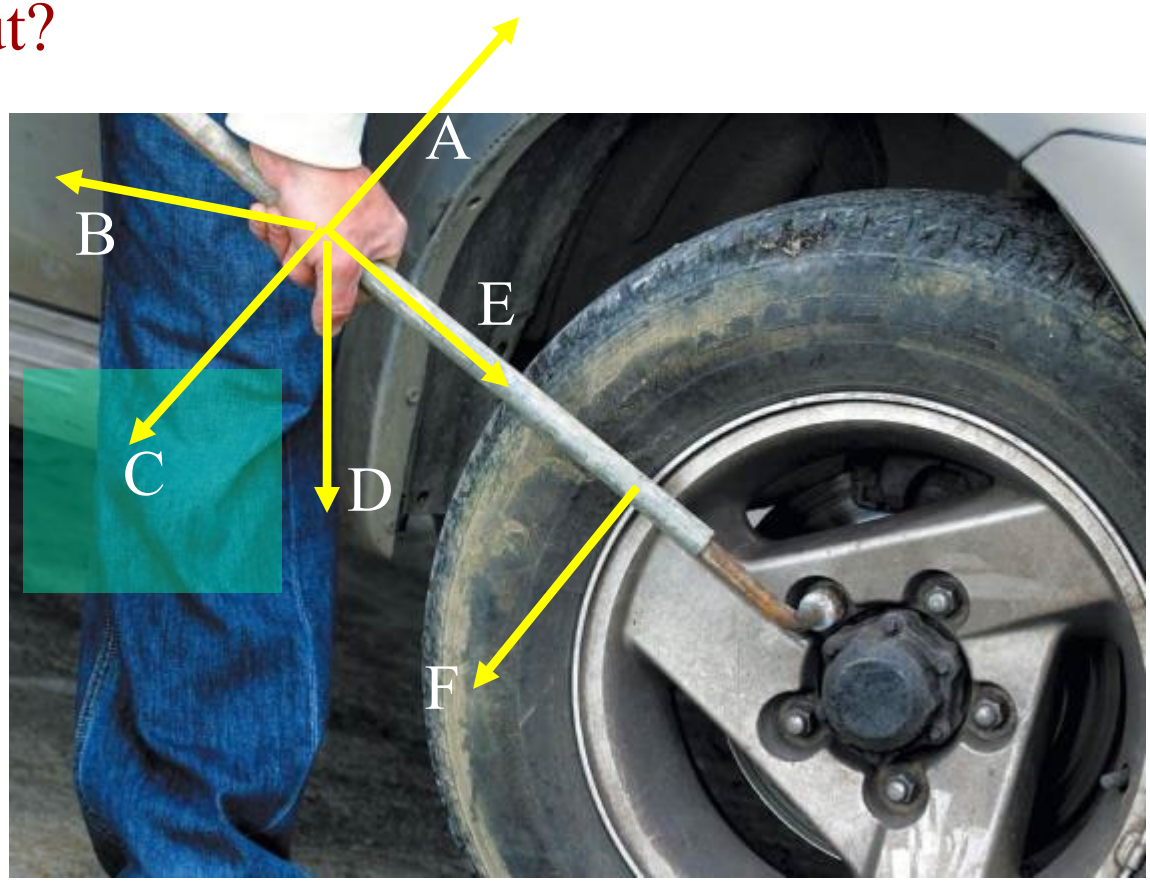
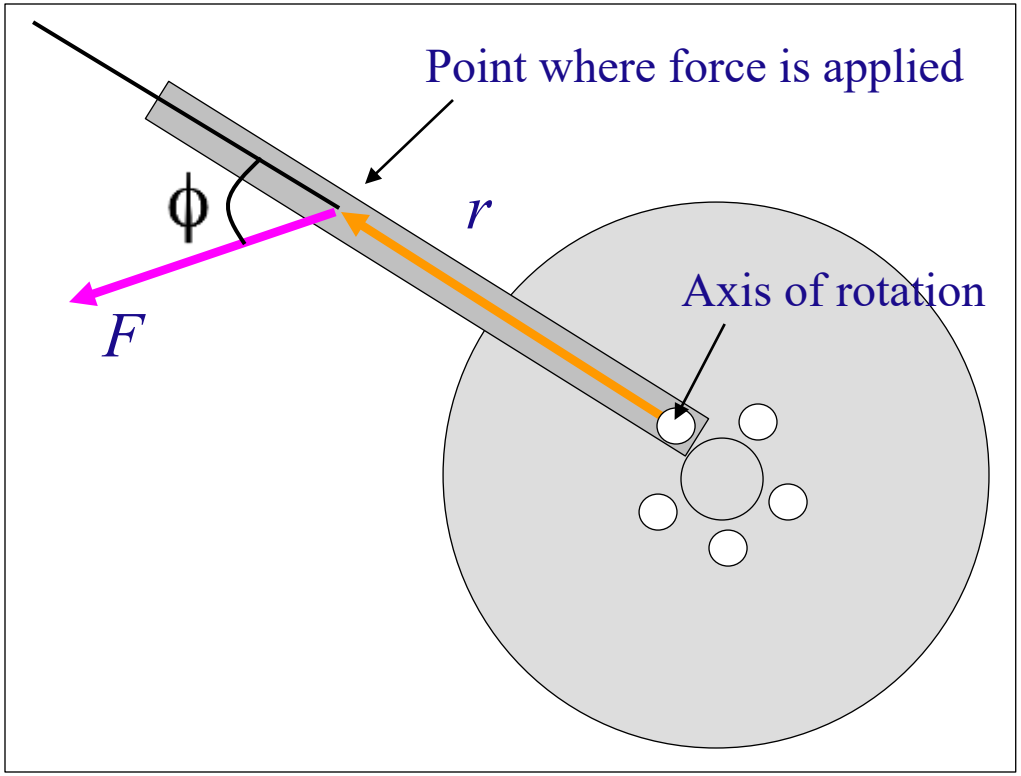
Question – SJWW 9.4,5

Which force would be most effective in loosening the wheel nut?



Torque - G 8.4

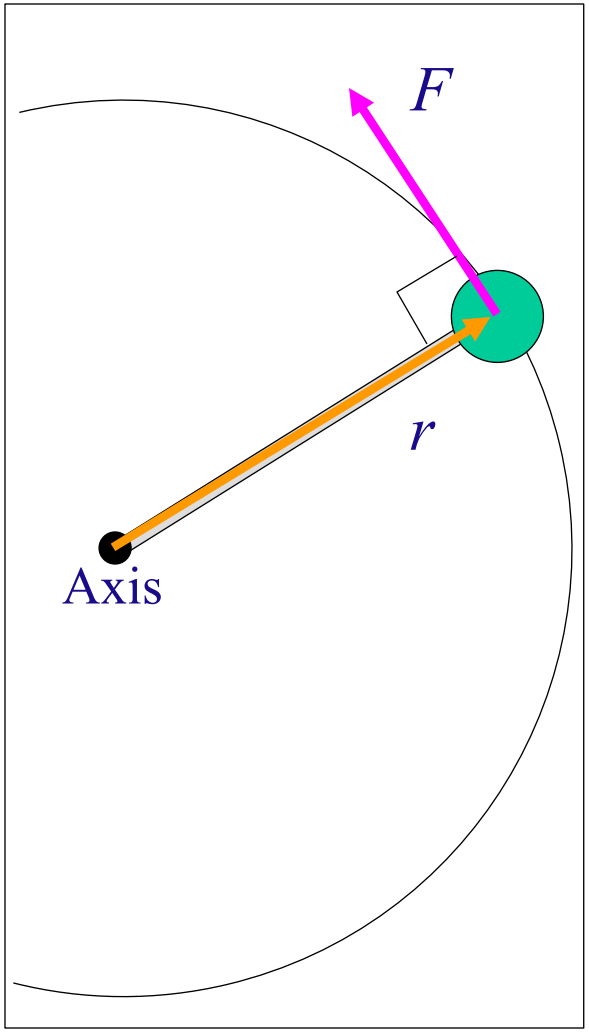


Definition of torque:

$$\tau \equiv rF\sin\phi$$

- Torque is the 'turning effect' of a force.
- Torque depends on
  - The *magnitude* of the force.
    - The *distance from the axis of rotation* to the point where the force is applied.
    - The *angle* between the force and a line drawn from the axis to the point where the force is applied.
- For a larger torque increase *r* or *F,* or both.
- For given values of *F* and *r*, the maximum possible torque occurs when $\phi = 90^\circ$.

Torque causes angular acceleration - G 8.5



Particle of mass *m* attached to a rigid, massless rod free to rotate about one end.

- Force *F* is applied to particle, perpendicular to the rod.

$$\tau = rF\sin\phi = rF$$

- Tangential acceleration of particle: $a_t = \dfrac{F}{m}$

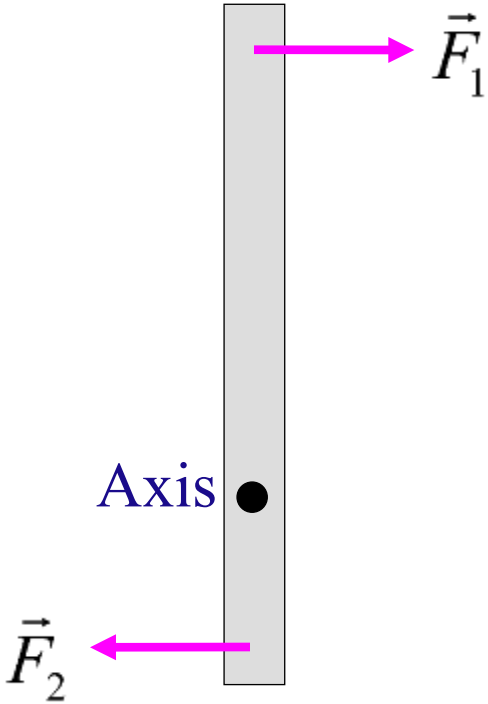
- Angular acceleration of particle:

$$a_t = \alpha r \qquad \text{so} \qquad \alpha = \frac{a_t}{r} = \frac{F}{mr}$$

$$\alpha = \frac{\tau}{mr^2}$$

Angular acceleration → $\alpha = \dfrac{\tau}{I}$ ← Torque, ← Moment of inertia

Static equilibrium - G 9.2



- Suppose that in the diagram on the left, we have $\vec{F}_1 = \vec{F}_2$

$$\vec{F}_{net} = \vec{F}_1 + \vec{F}_2 = 0 = m\vec{a}_{com}$$

- The net *force* on the object is zero and the object's centre of mass does not accelerate.
- But, the net *torque* on the object is *not* zero – the object has an angular acceleration (it rotates clockwise).

$$\vec{\tau}_{net} = I\vec{\alpha}$$

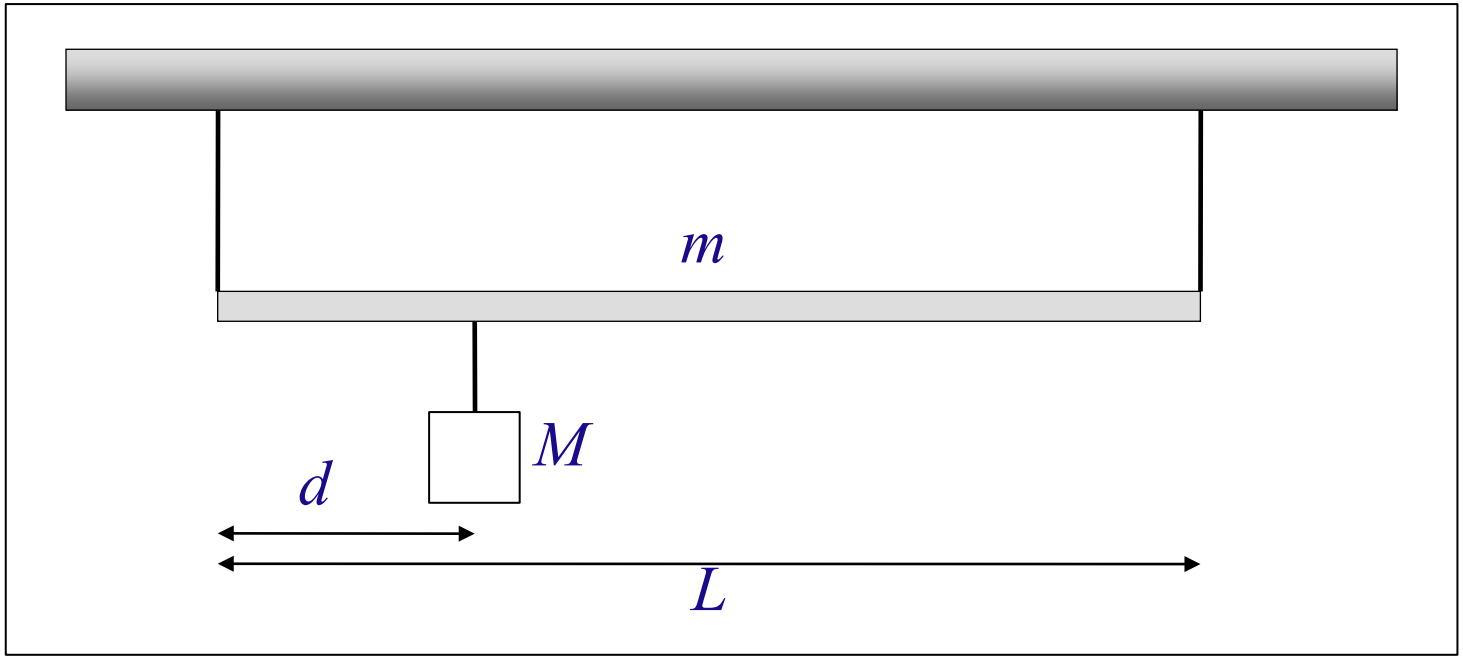
If an object is in static equilibrium:

- Net force is zero.
- Net torque about *any* axis is zero.

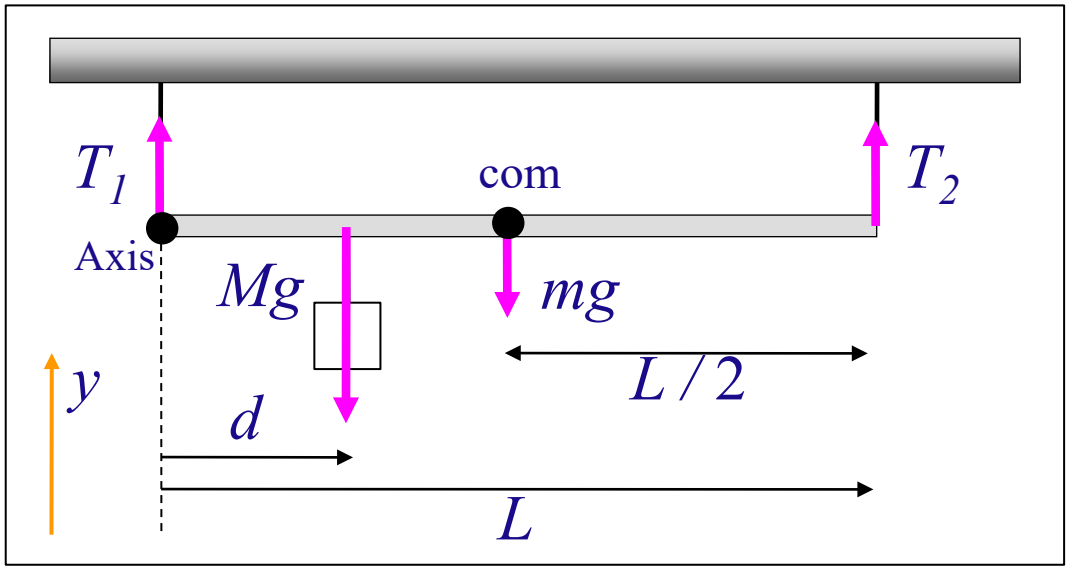
(MI -1 demo)

Milk carton mobile – G 9.2

A rod of mass m = 0.2 kg and length L = 1.0 m is suspended from the ceiling by strings tied to each end, as shown. If a milk carton of mass M = 1.0 kg is hung a distance d = 30 cm from one end of the rod, what are the tensions in each of the supporting strings?



Milk carton mobile – G 9.2



$m = 0.2$ kg, $L = 1.0$ m,
$M = 1.0$ kg, $d = 0.30$ m

$$\tau = rF\sin\phi$$

(We can choose *any* point as the axis.)

(Note the signs of the torques in this equation.)

Torques around left end of rod

$$-Mgd - mg\frac{L}{2} + T_2 L = 0 \qquad T_2 = \frac{Mgd}{L} + \frac{mg}{2} = 3.9\ \text{N}$$

Forces in y direction:

$$T_1 - Mg - mg + T_2 = 0 \qquad T_1 = Mg + mg - T_2 = 7.8\ \text{N}$$

D Moments demo